

Capacitor Charging

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Abstract

This document introduces the charging behaviour of a resistor-capacitor (RC) network and derives the capacitor voltage at one time constant.

1 Introduction

This document introduces the charging behavior of a resistor-capacitor (RC) network. When a capacitor C is charged through a series R from a constant supply V_0 , the voltage of the capacitor increases according to the condition.

$$V(t) = V_0 \left(1 - e^{-t/RC}\right), \quad (1)$$

where the product RC has units of seconds and is called the time constant $\tau = RC$.

2 Voltage at the Time Constant

Substituting $t = \tau$ into (1):

$$\begin{aligned} V(\tau) &= V_0 \left(1 - e^{-\tau/RC}\right) \\ &= V_0 \left(1 - e^{-RC/RC}\right) \quad \text{since } \tau = RC \\ &= V_0 (1 - e^{-1}) \\ &\approx V_0 (1 - 0.368) \\ &\approx 0.632 V_0. \end{aligned} \quad (2)$$

The exponent cancelling to -1 in (2) is why the 0.632 fraction is universal – R and C dropped out of the calculation before the final line, so this fraction holds for any RC network. Substituting $V_0 = 5.0 \text{ V}$:

$$V(\tau) \approx 0.632 \times 5.0 \text{ V} \approx 3.16 \text{ V}, \quad (3)$$

matching the dashed line in Figure 1.

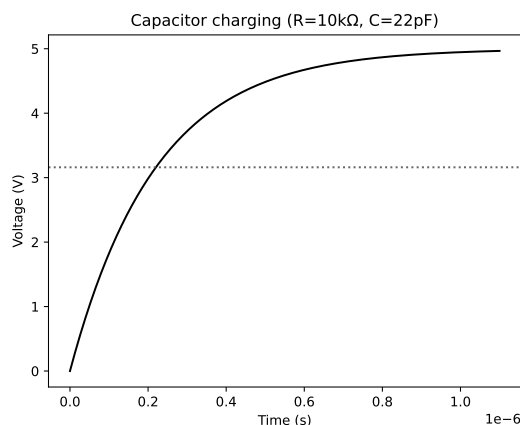


Figure 1: Charging curve for $R = 10 \text{ k}\Omega$, $C = 22 \text{ pF}$.